

ASPECT	DESCRIPTIVE STATISTICS	INFERENCEAL STATISTICS
• BASIC PURPOSE	Summarizes and describes main features of a dataset collected.	• Uses sample data to make inferences / prediction about a population.
• FOCUS	Deals with what the data show directly (FACTS about the sample / population data collected).	• Focuses on what goes beyond the data, estimating or testing claims about a wider population.
• TYPICAL TOOLS / MEASURES	Measures of central tendency (mean, median & mode), dispersion (range, variance, SD), Frequency tables, charts, graphs.	• Confidence Intervals, Hypothesis Test, Regression, ANOVA, Chi-Square Test, Correlation used for ESTIMATION & D.M.
• DATA SCOPE	Often describes a sample or sometimes an entire population, without extending conclusions beyond the observed data.	• Always starts from a sample and generalizes result to the whole population using probability theory.
• TYPES OF CONCLUSION	Provides descriptive summaries (e.g., "average score is 75").	• Provides probabilistic statements and decisions (e.g., "there is evidence that true mean is greater than 70").
• USE STAGE IN ANALYSIS	Usually the first step in data analysis to <u>understand</u> and <u>clean</u> the data.	• follows descriptive analysis to formally <u>test Hypothesis</u> or <u>build predictive models</u> .